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1) a) Simple Monthly Return

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$R_1 = \frac{108 - 100}{100} = 0.08$$

$$R_2 = \frac{103 - 108}{108} = -0.0463$$

$$R_3 = \frac{115 - 103}{103} = 0.1165$$

$$R_4 = \frac{110 - 115}{115} = -0.0435$$

b) log Return: $\pi_t = \ln(P_t / P_{t-1})$

$$\pi_1 = \ln\left(\frac{108}{100}\right) = 0.077$$

$$\pi_2 = \ln\left(\frac{103}{108}\right) = -0.0413$$

$$\pi_3 = \ln\left(\frac{115}{103}\right) = 0.110 \quad \pi_4 = \ln\left(\frac{110}{115}\right) = -0.044$$

c) ~~Month~~ verification of the approximation $\pi_t \approx R_t$

Month	R_t	π_t	$ R_t - \pi_t $
1	0.08	0.077	0.0030
2	-0.0463	-0.0413	0.0010
3	0.1165	0.1100	0.0065
4	-0.0435	-0.0444	0.0009

The approximation is least accurate in Month 3, as the return is the largest in magnitude ($\approx 11.65\%$). The relation $r \approx R$ comes from the first order Taylor expansion $\ln(1+R) \approx R$ which works well only if R is very small. For larger R , the higher order terms matter and the gap grows.

② (a) Linearity of expectation says: for constant w_1, w_2 .

$$E[w_1 R_1 + w_2 R_2] = w_1 E[R_1] + w_2 E[R_2]$$

Applying it here:

$$E[R_p] = 0.4 \times 15\% + 0.6 \times 7\% = 60.2\%$$

(b) No result do not change. Linearity of expectation always holds, regardless of how R_1 & R_2 are correlated. Correlation affects the variance of the portfolio, not its expected return.

③
$$\sigma_p^2 = \text{var}(w_1 R_1 + w_2 R_2)$$

Apply $\text{var}(x+y) = \text{var}(x) + \text{var}(y) + 2 \text{cov}(x, y)$

$$\sigma_p^2 = \text{var}(w_1 R_1) + \text{var}(w_2 R_2) + 2 \text{cov}(w_1 R_1, w_2 R_2)$$

using $\text{var}(cx) = c^2 \text{var}(x)$ & $\text{cov}(ax, by) = ab \text{cov}(x, y)$

$$= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \text{cov}(R_1, R_2)$$

Substitute $\text{cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

④ Given, $\mu_A = 18\%$, $\sigma_A = 25\%$, $\mu_B = 8\%$, $\sigma_B = 12\%$, $w_A = w_B = 0.5$

The portfolio variance becomes:

$$\sigma_P^2 = (0.5)^2(25)^2 + (0.5)^2(12)^2 + 2(0.5)(0.5)\rho(25)(12)$$

$$= 156.2 + 36 + 150\rho$$

① $\rho = +1$, $\sigma_P^2 = 342.25 \Rightarrow \sigma_P = 18.5\%$

$\rho = 0$: $\sigma_P^2 = 192.25 \Rightarrow \sigma_P = 13.87\%$

$\rho = -1$ $\sigma_P^2 = 42.25 \Rightarrow \sigma_P = 6.5\%$

⑤ σ_P equals the weighted average when $\rho = +1$

Proof: When $\rho = 1$

$$\sigma_P = \sqrt{w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B}$$

$$= \sqrt{(w_A \sigma_A + w_B \sigma_B)^2} = w_A \sigma_A + w_B \sigma_B$$

Numerically = $(0.5)(25) + 0.5 \times 12 = 18.5\%$

⑥ for $\rho = -1$ setting $\sigma_P = 0$

$$w_A^* \sigma_A = (1 - w_A^*) \sigma_B \Rightarrow w_A^*(25) = (1 - w_A^*)(12)$$

$$\rightarrow 37w_A^* = 12$$

$$\therefore w_A^* = \frac{12}{37} \approx 0.324$$

⑤ The portfolio variance formula is

$$\sigma_P^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

When $\rho_{12} < 1$, the cross-term is smaller compared to the case $\rho_{12} = 1$. This makes σ_p^2 fall below $(w_1\sigma_1 + w_2\sigma_2)^2$, so portfolio risk is less than the weighted average of individual risks. Intuitively, the two assets do not move together, so their ups & downs partially cancel out.

The benefit disappears only when $\rho_{12} = 1$ that is perfect comovement.

⑥ a) using $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$

$$\Sigma_{11} = (0.25)^2 = 0.0625$$

$$\Sigma_{22} = (0.12)^2 = 0.0144$$

$$\Sigma_{33} = (0.04)^2 = 0.0016$$

$$\Sigma_{12} = \Sigma_{21} = 0.4 \times 0.25 \times 0.12 = 0.0120$$

$$\Sigma_{13} = \Sigma_{31} = 0.1 \times 0.25 \times 0.04 = 0.0010$$

$$\Sigma_{23} = \Sigma_{32} = 0.2 \times 0.12 \times 0.04 = 0.00096$$

$$\Sigma = \begin{bmatrix} 0.0625 & 0.0120 & 0.0010 \\ 0.0120 & 0.0144 & 0.00096 \\ 0.0010 & 0.00096 & 0.0016 \end{bmatrix}$$

⑥ The matrix is symmetric because $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j = \rho_{ji}\sigma_j\sigma_i = \Sigma_{ji}$

⑦ Sharpe ratio = $\frac{E[R_p] - R_f}{\sigma_p}$ with $R_f = 4\%$.

$$S_X = \frac{14 - 4}{22} \approx 0.455$$

$$S_Y = \frac{9 - 4}{10} = 0.5$$

$$S_Z = \frac{20 - 4}{35} = 0.457$$

⑥ Ranking $Y > Z > X$

A rational mean variance investor maximizes return per unit of risk. Portfolio Y has the highest Sharpe Ratio, so it is the most preferred.

⑦ Portfolio Z looks attractive at first because its return 20% is the highest. But it also has the most risk ($\sigma = 35\%$)

↳ its Sharpe Ratio (0.451) is actually lower than Y's (0.5). So Y gives more return for every unit of risk taken. A rational investor cares about this trade off, not just the raw numbers.

⑧

a) The Global minimum Variance Portfolio is the combination of assets that achieves the lowest possible variance across the entire feasible set. It sits at the leftmost tip of the minimum variance frontier in $(\sigma_p, E[R_p])$ space, because no other feasible portfolio can have smaller risk.

b) The student is incorrect. Every portfolio on the lower half of the minimum variance frontier is dominated, there exists another portfolio with the same σ_p but a higher $E[R_p]$ on the upper half. No rational investor accepts a lower return for the same level of risk. A very risk-averse investor should hold the GMV portfolio.

c) The CML is the straight line drawn from the risk free rate $(0, R_f)$ tangent to the efficient frontier. The point of tangency is the market portfolio M.

$$\text{Slope of CML is } = \frac{E[R_M] - R_f}{\sigma_M}$$

③ a) with $n = 86$, observations & $\sigma_i = 30\%$.

$$SE(R_i) = \frac{\sigma_i}{\sqrt{n}} = \frac{0.3}{\sqrt{86}} = 5\%$$

This is large uncertainty. expected return for most assets lie roughly in 8-15% range is $\pm 5\%$. even make historical estimate quite unreliable.

④ b) MVO is called an error maximiser because it tends to assign large weights to assets with overestimated returns caused by noisy data, which amplifies estimation error. Using ML-based return prediction provides another & more forward looking estimator, helping reduce overfitting & making portfolio allocation more stable.

⑤ c) For N assets, Σ has

$$\begin{aligned} \text{unique parameters} &= \underbrace{N}_{\text{var}} + \underbrace{\frac{N(N-1)}{2}}_{\text{covar.}} = \frac{N(N+1)}{2} \end{aligned}$$

$$N = 80 \quad \Rightarrow \quad \frac{80 \times 81}{2} = 1275$$

That is so variations & 1275 unique covariances. Estimating all 1275 parameters accurately need a large amount of data. With limited history the sample covariance matrix becomes unreliable & can be even be singular, causing serious numerical problems in optimisation.